

DHANA KARTHIKEYA VENTRAPRAGADA

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EDUCATION

University of California, Berkeley

Expected Dec 2027

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00

Berkeley, CA

- **Relevant Coursework:** Data Structures · Structure & Interpretation of Computer Programs · Machine Learning · Data Engineering · Efficient Algorithms and Intractable Problems · Principles & Techniques of Data Science · Probability for Data Science

TECHNICAL SKILLS

Languages: Python, SQL, Java, Scheme

ML & Data Science: pandas, NumPy, scikit-learn, PyTorch, XGBoost, LightGBM, SentenceTransformers

Backend & Infrastructure: FastAPI, Pydantic, PostgreSQL, Auth0, AWS S3, Gemini API, ffmpeg, React

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook

EXPERIENCE

Guang Labs | Engineering Intern | Remote

May 2026 – Aug 2026

- Built an async video/image domain-classification API (FastAPI, Auth0, Postgres job store): submit-and-poll REST endpoints run classification as background jobs with persisted status and granular per-item progress.
- Implemented the Python pipeline end-to-end: resolving media from 4 source types (local, S3, HTTP), auto-detecting image vs. short/long video via fprobe/ffmpeg, and classifying via Gemini multimodal calls with a trust-gated resample loop.
- Split the service into isolated, testable modules behind a central orchestrator, backed by 86 unit, integration, and acceptance tests; separately built an end-to-end in-app notification framework (Postgres, backend APIs, a React dashboard bell) wired into the ingest and export pipelines.

IDX Exchange | Data Science Intern | Remote

Jun 2026 – Aug 2026

- Built a preprocessing pipeline for a California home-price model on 70K+ CRMLS single-family records — data cleaning, statistical outlier caps, and a chronological train/test split holding out the latest month.
- Diagnosed and fixed a train/test data-leakage bug in the outlier-threshold logic, recomputing thresholds from training data only; benchmarked four models, with the final XGBoost model reaching R^2 0.88 and MdAPE 8.64% on the held-out month.

PROJECTS

Movie Genre Classification (Multi-Label NLP) | Python, PyTorch, SentenceTransformers

- Built a multi-label genre prediction system on 42K+ MovieLens/TMDB films, parsing nested IMDb metadata with regex, engineering log budget/revenue and temporal features, and embedding title+tagline+overview text via MiniLM, MPNet, and e5-large-v2.
- Benchmarked Logistic Regression, XGBoost, and LightGBM on 1024-dim e5 embeddings; combined a rare-genre cutoff with threshold tuning to address class imbalance, lifting Macro F1 from 0.48 to 0.66 (Micro F1 0.68) with Logistic Regression beating tree models while training $\sim 12\times$ faster.

Email Spam/Ham Classifier | Python, scikit-learn, pandas, NumPy

- Engineered 50+ features from raw emails — regex counts (links, punctuation, dollar signs), HTML/subject indicators, log-scaled length features, and a curated spam/ham bag-of-words.
- Trained L1-regularized Logistic Regression tuned via GridSearchCV and validated with 10-fold cross-validation, achieving 92.1% test accuracy on the held-out leaderboard set.

LEADERSHIP & INVOLVEMENT

UC Berkeley CDSS | Reader/Tutor, CDSS 198 | Berkeley, CA

Aug 2026 – Present

- Facilitate weekly seminar discussions for CDSS 198: Careers in Data Science & Computing, giving feedback on student resumes, cover letters, and development plans and distributing job/internship listings; separately support transfer students in CDSS 198: Transfer Seminar across two weekly class sessions, office hours, and on-demand tutoring.

UC Berkeley EECS | UCS1, CS 61A | Berkeley, CA

Jun 2026 – Aug 2026

- Ran 4 weekly exam-prep sections and office hours for CS 61A (Python, SQL, Scheme), built autograders on Gradescope, handled regrade/extension requests, and wrote and reviewed midterm and final exam questions.

Boundary.0, De Anza College | Secretary | Cupertino, CA

Apr 2024 – Jun 2025

- Helped grow a new student organization to 120+ members as one of its first officers, partnering with 5+ companies across construction, finance, and computer science to connect students with hands-on industry experience.